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## **Efficient Fluid Sampling and Comprehensive Fast-Track Geological Evaluation for Oil Potential in Lower Cretaceous Carbonate Reservoirs, Abu Dhabi Offshore**

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### **Abstract**

As part of ADNOC's strategic priorities and the Upstream 2.0 initiative, accelerating the development of undeveloped reservoirs is essential to future-proof the business and respond to evolving global energy demands. In this context, there was a recognized necessity to pursue new hydrocarbon opportunities within our offshore fields by leveraging the latest advancements in oil and gas industry technologies—specifically, cutting-edge methods for fluid sampling in tight reservoirs. This approach was complemented by implementing a fast-track modeling workflow, which enabled the rapid evaluation of potential hydrocarbon volumes in place. Alongside this, robust uncertainty analysis was conducted to help inform decision-making and optimize asset development strategies.

Acquiring representative fluid samples from tight and challenging reservoir environments has long posed significant technical difficulties due to their low permeability and complex rock properties. To overcome these challenges, advanced reservoir penetration techniques and the deployment of industry-leading sampling tools were employed. These specialized devices allowed for successful extraction of fluid samples in conditions that would traditionally impede accurate data collection, ensuring the integrity and reliability of the samples gathered. This process plays a pivotal role in the precise evaluation of the reservoir, facilitating accurate identification of hydrocarbon potential and enabling informed field development planning.

Upon completion of the well interventions and data acquisition campaigns, a rapid two-dimensional analysis was performed, focusing on the lateral variation of reservoir properties and the estimation of possible Free Water Levels (FWL) across multiple wells. This information served as the foundation for the construction of a fast-track static reservoir model. Within this model, Oil-Water Contacts (OWC) and various reservoir rock typing scenarios were postulated, which were subsequently incorporated into a comprehensive sensitivity analysis to estimate a range of Oil Initially In Place (OIIP) volumes.

One of the significant achievements of this project was the first-time deployment of state-of-the-art technology for pressure and reservoir fluid sampling using a multi-probe system with an exceptionally large

area packer. Unlike conventional methods that often necessitate the use of inflatable packers in low-mobility formations, this innovative tool enabled efficient testing and sampling without additional equipment, ultimately streamlining operations and reducing risk. The successful application of this advanced technology not only provided high-quality fluid samples from a previously unsampled tight reservoir but also set a new benchmark for future reservoir evaluation efforts.

Following the successful data acquisition phase, two-dimensional maps displaying porosity, permeability, and net pay thickness were generated using input from all available well data. These visualizations enhanced the understanding of reservoir trends and heterogeneities. Further, several rock typing methodologies, including the Winland and Lucia models, were evaluated to assign reservoir rock types and assess their impact on volumetric estimations. Minimum, mid, and maximum Free Water Level (FWL) values were derived from well logs in the newly appraised reservoir, feeding into sensitivity analyses to better understand their influence on volumetric calculations and resource estimates.

This abstract highlights the practical application of a breakthrough fluid sampling tool enabling the acquisition of representative fluid samples from tight reservoirs for the first time. It also demonstrates the value of integrating fast-track data analysis and modeling within a comprehensive workflow to improve understanding of reservoir characteristics, Free Water Level variations, and their collective impact on volumetric assessment. The lessons learned and methods developed through this work pave the way for more effective reservoir evaluation and optimized resource development in challenging environments, supporting ADNOC's continued commitment to innovation and operational excellence.

## Introduction

In the realm of petroleum exploration and production, the relentless pursuit to maximize hydrocarbon recovery and mitigate field development risks remains at the forefront of strategic priorities. One of the most critical elements in this endeavor is the judicious placement of appraisal wells, particularly on the flanks of geological structures. While discovery wells serve the vital role of confirming the presence of hydrocarbons, it is the meticulous planning and drilling of subsequent appraisal wells—especially those positioned on the structural peripheries—that truly refine and deepen our understanding of the reservoir's architecture, its lateral and vertical extents, and its overall hydrocarbon potential.

The significance of these appraisal wells cannot be overstated. By targeting the flanks, operators are able to delineate the boundaries and heterogeneities of the reservoir, identify variations in rock and fluid properties, and reduce the uncertainties associated with volumetric estimates. This detailed structural and petrophysical mapping lays the groundwork for more accurate modeling, improved well placement, and ultimately, optimal field development strategies. In the context of new appraisal wells, unlocking previously untapped hydrocarbon potential becomes a pivotal factor in meeting the world's continually evolving energy demands and in maximizing the return on investment across the full lifecycle of a field.

Amidst this broader landscape, the discovery and efficient exploitation of tight, minor reservoirs—characterized by their intrinsically low permeability and often limited areal distribution—have grown in both strategic and economic importance. Although minor reservoirs may be overlooked in favor of larger, more prolific fields, technological advancements and changing market conditions have increasingly shifted focus towards these challenging, yet promising, accumulations. These reservoirs typically present unique development challenges: their confined thickness, narrow areal footprint, and sometimes complex compartmentalization can impede both evaluation and production. Yet, these characteristics also mean minor reservoirs are often nestled within or adjacent to mature producing areas, where infrastructure exists and incremental production can be rapidly realized.

Examples of such reservoirs include low-permeability carbonate formations with intricate and tight pore networks, or thin, highly compartmentalized stratigraphic intervals that demand precision in both data gathering and modeling. Through targeted exploration and advanced reservoir characterization techniques,

these smaller resource pools can be systematically assessed, unlocking their value and extending the productive life of mature oil provinces.

This paper delves into how a lower Cretaceous, minor, tight, and thin carbonate reservoir opportunity was not only identified but strategically maximized through careful planning and execution of a comprehensive data acquisition program. By selecting and deploying the most suitable acquisition tools, the project team ensured that critical reservoir data—previously unattainable—were successfully captured. This meticulous approach led, for the first time in the field's history, to the successful flow of hydrocarbons from this minor reservoir, marking a significant technical milestone.

Furthermore, the study illustrates how an integrated, fast-track geological evaluation was employed to rapidly interpret the gathered data, identify reservoir trends, and delineate hydrocarbon-water contacts (HWCs). This approach enabled the swift construction of a simple yet robust geological model, facilitating volumetric estimations across a range of uncertainty scenarios. Through rigorous sensitivity and uncertainty analyses, the team was able to better quantify resource potential and inform strategic decisions for future development.

Ultimately, the lessons gleaned from this endeavor not only underscore the value of precise data acquisition and fast-track evaluation in challenging environments but also highlight a pathway toward more efficient and effective resource development strategies. The methodologies and insights established through this work can be adapted and applied to similar minor reservoir opportunities, supporting the broader goal of sustaining and enhancing energy production in a rapidly changing global landscape.

## Objectives

The objective of this paper is to present in detail the significant added value brought by deploying the latest-generation fluid sampling tools within the context of the oil and gas industry, specifically as it pertains to unlocking the potential of minor and technically challenging reservoirs. By leveraging these cutting-edge technologies, operators are now able to obtain higher-fidelity samples that accurately reflect the complex fluid composition and dynamic characteristics present within tight, thin carbonate intervals that have historically eluded comprehensive evaluation. This advancement is particularly critical in reservoirs where conventional sampling methods frequently fall short due to low permeability, narrow pay zones, and high compartmentalization, all of which can mask the true hydrocarbon potential and impede sound resource assessment.

In addition to highlighting the technological leap represented by modern fluid sampling tools, this paper systematically outlines the comprehensive methodology adopted for evaluating a newly identified minor reservoir. The evaluation process integrates both two-dimensional (2D) and three-dimensional (3D) modeling techniques, ensuring a holistic approach that bridges the gap between raw data collection and actionable reservoir characterization. The methodology begins with the meticulous acquisition of core and well-log data, followed by the application of advanced petrophysical interpretation workflows designed to unravel the subtle heterogeneities and intricate pore structures typical of these reservoirs. Subsequently, the workflow transitions into the construction of detailed 2D geological cross-sections and, ultimately, the development of 3D reservoir models that encapsulate both lateral and vertical complexities.

Crucially, the study places a strong emphasis on the rigorous treatment of uncertainties at every stage of the evaluation. By employing stochastic modeling, sensitivity analyses, and scenario planning, the team systematically assesses the potential impacts of key geological and petrophysical variables on volumetric calculations. Parameters such as porosity, permeability, water saturation, reservoir thickness, and structural configuration are all subjected to robust uncertainty analysis, providing a spectrum of possible outcomes rather than a single deterministic estimate. This approach not only enhances the reliability of volumetric estimations but also empowers decision-makers with a clearer understanding of risk and potential upside, facilitating more informed strategic choices regarding further appraisal, development, and investment.

Ultimately, the findings presented in this paper underscore the transformative potential of integrating state-of-the-art fluid sampling technology with sophisticated geological evaluation methods. The lessons learned from this case study are broadly applicable, offering a blueprint for efficiently and effectively evaluating similar minor reservoir opportunities in other mature basins around the world. In doing so, this work contributes toward optimizing resource recovery, maximizing economic returns, and supporting the ongoing evolution of the energy sector in response to growing global demands.

## Data acquisition

### Wireline Logs acquisition

Full set of Wireline Logs was run in the well covering the studied minor reservoir as shown in below [Table 1](#):

**Table 1—showing all the wireline logs which was run in the well covering the studied reservoir**

| Logs                  |                            | Studied reservoir |
|-----------------------|----------------------------|-------------------|
| Geology               | Image                      | √                 |
| Formation Evaluation  | Triple-Combo               | √                 |
|                       | NMR                        | √                 |
|                       | Dielectric                 | √                 |
| Seismic               | Sonic                      | √                 |
|                       | ZOVSP from surface (Cont.) | √                 |
| Reservoir Engineering | Pre-tests                  | √                 |
|                       | FID/Sampling               | √                 |

### Petrophysical Interpretation

The studied reservoir had not previously been tested in earlier wells, largely due to its tight nature and high water saturation. Nevertheless, in this appraisal pilot, the initial mud logging results did not indicate significant amounts of gas; yet, with each subsequent data acquisition run, subtle indications of potential within the reservoir began to emerge. Petrophysical analysis presented promising findings: the reservoir exhibited an average porosity of 15 percent over a thickness of 9 feet, permeability measured at 5 millidarcies, and oil saturation reached 60 percent, as illustrated in [Figure 2b](#). Dielectric measurements further highlighted the presence of residual oil, especially concentrated near the upper section of the reservoir, reinforcing the prospect of hydrocarbons.

Following these encouraging signs, formation tester runs were performed with notable success. Three valid pretests were conducted, each yielding a pressure reading of 4580 psi. During fluid identification, oil breakthrough was observed almost immediately; the oil fraction reached up to 60 percent, and the maximum water salinity recorded was 90,000 ppm—significantly below the known formation water salinity, further suggesting the presence of mobile hydrocarbons. In total, three bottles of fluid samples were collected for further laboratory analysis, providing critical data for compositional and PVT studies. Based on these findings, a 12-foot interval was carefully selected and designed for perforation to carry out a comprehensive Drill Stem Test (DST) analysis.

This integrated workflow, combining advanced petrophysical evaluation, robust fluid sampling, and thorough pressure testing, has shed new light on the reservoir's potential. The results not only validate the methodology adopted but also offer actionable insights for future appraisal and development, paving the way for maximizing resource recovery in challenging reservoir environments.



## Innovative new Fluid Sampling technology

New technology that was first introduced in the field and deployed in the new drilled appraisal well, The capabilities of the tool, include a large area multi-probe sampling service with a total probe flow area of 66 square inches. The tool is designed to enable testing and sampling in ultra-low mobility reservoirs, with a broad mobility range extending to 0.1 md/cp. It is suitable for unconsolidated formations, thinly laminated formations, heavy oil sampling, and challenging boreholes. The slide emphasizes the safe and efficient deployment of the tool, which reduces the dependency on inflatable packers.

The results of the fluid sampling job was very positive as it enabled to collect Hydrocarbon sample for the first time in the field in this minor reservoir with reservoir properties of 14% porosity and permeability of 1-2 mD.

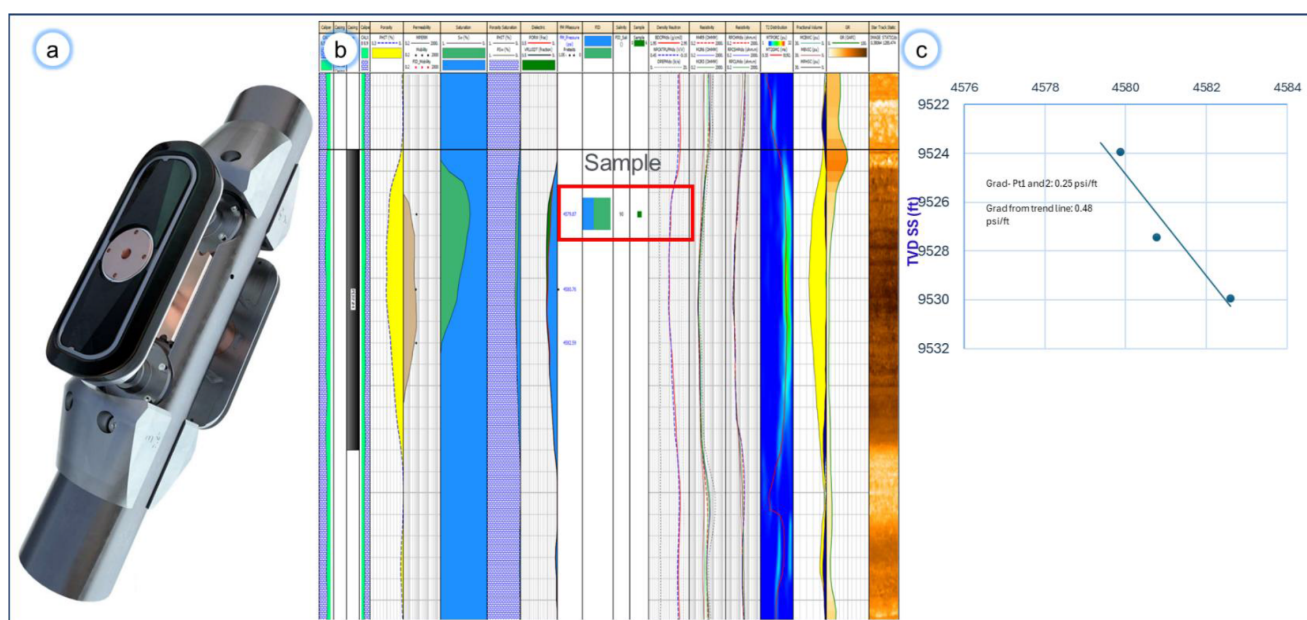


Figure 1—Sketch for the used fluid sampling tool and results: (a) shows the used tool with the probe size 66"; (b) results of the FID with the petrophysical interpretation; (c) Pressure vs Depth gradient for the minor reservoir.

## Fast-track Geological evaluation

### Available data

- 39-penetrations in field for the studied reservoir with full set of logs in wireline and LWD mode
- 3 Cores in 3 Appraisal and Pilot wells.

### Reservoir Overview

The Studied Minor reservoir considered to be tight reservoir which avg. Porosity Ranges from 9-13% pu and avg perm ranges from ~1 to 7 md. Avg. Gross Thickness of 14 ft and Avg Net Thickness is ~12 ft considering Porosity Cutoff 5%.

Depth structure map for the top of the reservoir is shown in Fig.2 and Gross and Net Thickness shown in Fig.3



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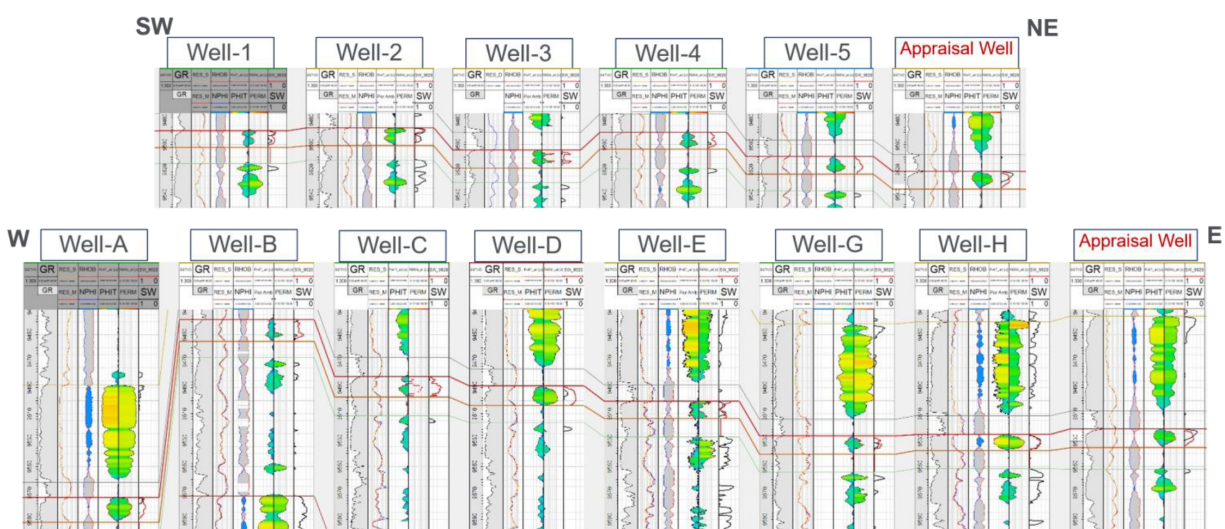


Figure 4—Cross Sections across reservoir in SW-NE for top X-Section, and W-E for Bottom X-Section

### HWC evaluation

Building upon the insights gleaned from offset reservoirs situated both above and below the studied interval, it became evident that the Free Water Level (FWL) exhibits distinct variations across different reservoir segments, largely due to compartmentalization by faults. This understanding led to the assumption that the studied reservoir would likely conform to a similar pattern of segmentation, as depicted in Figure 5a. To address these complexities, individual segments were assessed separately using data from existing wells, allowing for a more precise estimation of the most probable FWL for each compartment. These estimations were carefully aligned with water saturation (Sw) logs to ensure consistency and accuracy.

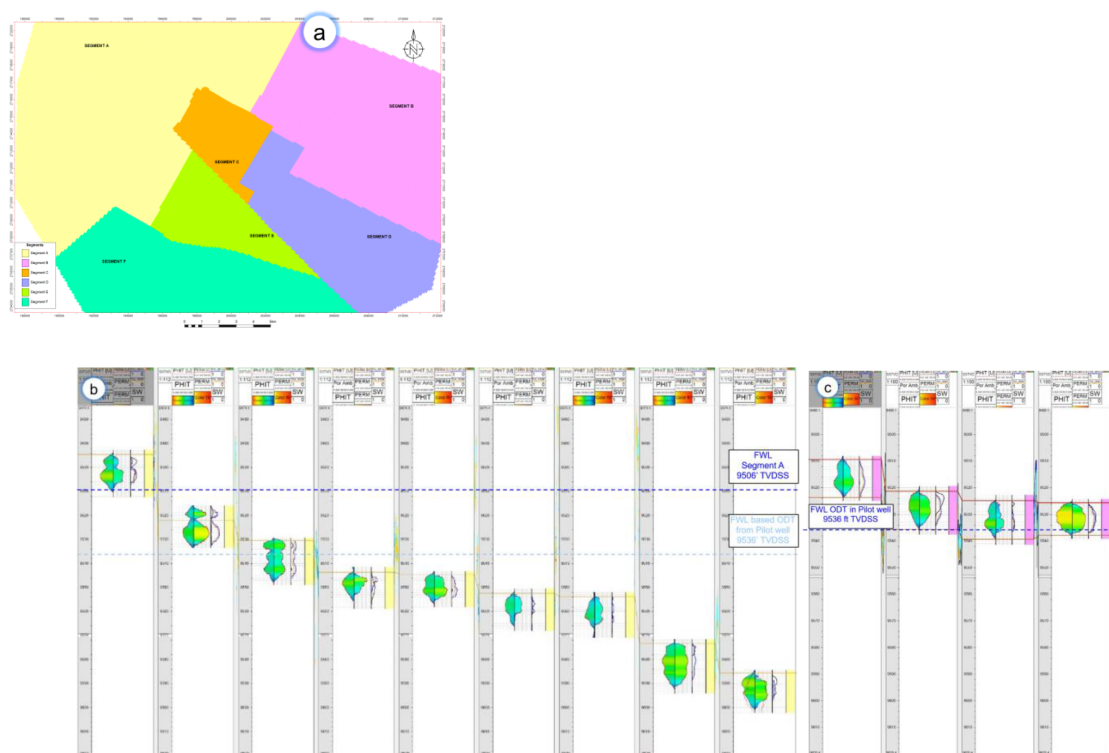


Figure 5—FWL evaluation (a) shows the different segments for FWL; (b) Segment A FWL analysis; (c) Segment B FWL analysis



An example of this segmentation and analysis exercise is detailed in [Figures 5b](#) and [5c](#), where the methodology for differentiating compartments, calibrating FWL, and integrating log data is illustrated. This stepwise approach not only facilitated a more granular understanding of the reservoir's hydrocarbon distribution but also helped inform future well placement and resource management strategies. By systematically dissecting each segment and correlating findings across multiple data sources, the team advanced the overall reservoir characterization, laying a robust foundation for ongoing appraisal and development activities

## Rock Typing

Fast Track approach was used to define Rock types for the wells penetrating the studied reservoir, to be used later on in the property Distribution, Two well Known approach was used Lucia and Winland, Rok type was defined at log level to be used in Property modeling while Static Modeling phase, below [figure 6a](#) and [6b](#) shows the poro-Perm plot colored by Rock types.

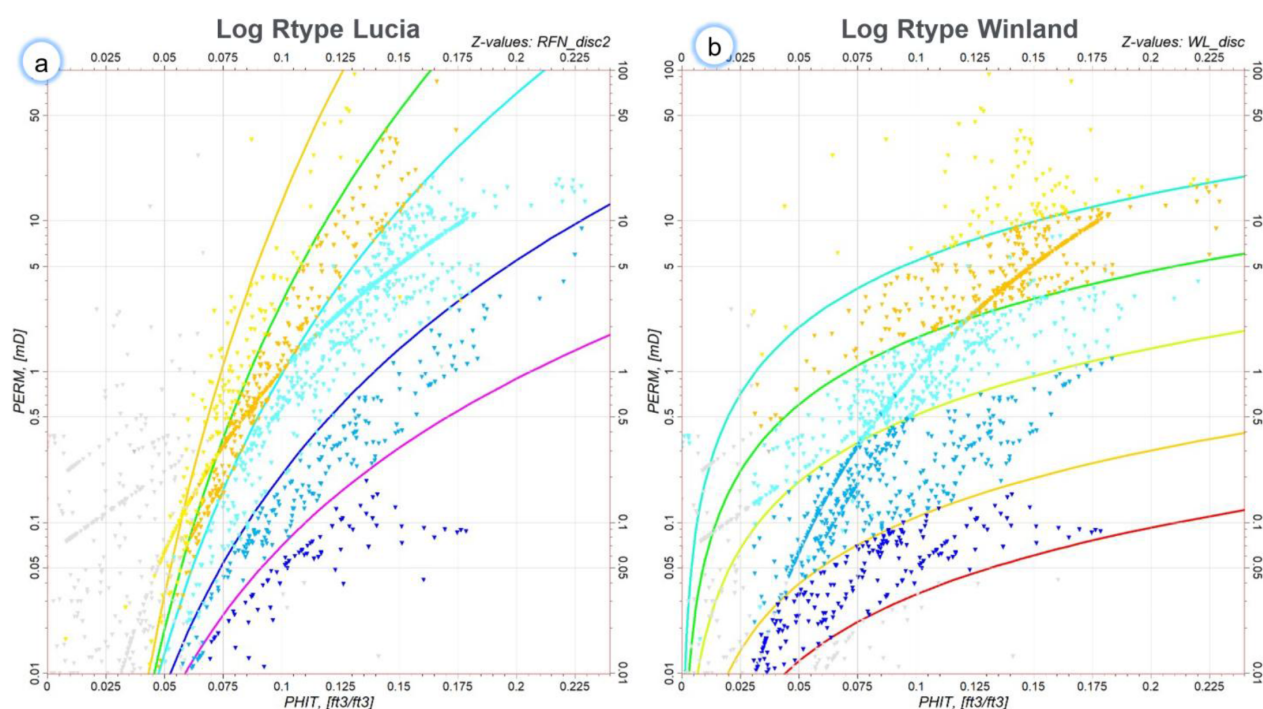


Figure 6—Poro Perm Plot for logs in studied reservoir colored by RT; (a) Lucia methodology; (b) Winland Methodology.

## Geological Modeling

The quick analysis required for evaluating the potential of the studied reservoir necessitated a well-structured and methodical geological modeling process. Initially, this process began with the careful identification and picking of the reservoir's top and base, followed by a thorough quality control (QC) to ensure the accuracy of the picked horizons. Building upon this foundation, 2D mapping was implemented to discern structural and stratigraphic trends, providing a crucial understanding of the spatial distribution and connectivity of reservoir facies. This stage also included the evaluation of the Free Water Level (FWL), which is vital for delineating hydrocarbon-bearing zones and for accurate volumetric calculations.

Subsequently, a detailed rock typing exercise was undertaken, incorporating various methodologies to characterize the reservoir rock, including both Lucia and Winland approaches. This was paired with property modeling, in which petrophysical properties such as porosity and permeability were modeled through multiple realizations. The process involved running scenarios with and without rock type constraints, and different trends were tested to account for spatial variability and geological heterogeneity within

the reservoir. The dual application of rock typing methodologies allowed for a more comprehensive understanding of reservoir quality and flow characteristics, ultimately feeding into improved property models.

The culmination of these multi-scenario workflows was the generation of a range of volumetric estimates, which were primarily influenced by uncertainties in property distribution. Sensitivity analyses assessed how variations in porosity, permeability, reservoir thickness, and spatial trends directly impacted the calculated hydrocarbon volumes. By systematically exploring these uncertainty scenarios, the team was able to present a robust spectrum of potential outcomes—rather than a single deterministic value—offering greater confidence in the subsequent resource assessment and development planning.

This integrated approach, encompassing detailed geological interpretation, advanced petrophysical modeling, and rigorous uncertainty analysis, exemplifies the best practices for evaluating complex, minor reservoirs. It not only maximizes the extraction of value from challenging reservoir settings but also facilitates more informed decision-making by quantifying risks and highlighting potential upside opportunities for future appraisal and development.

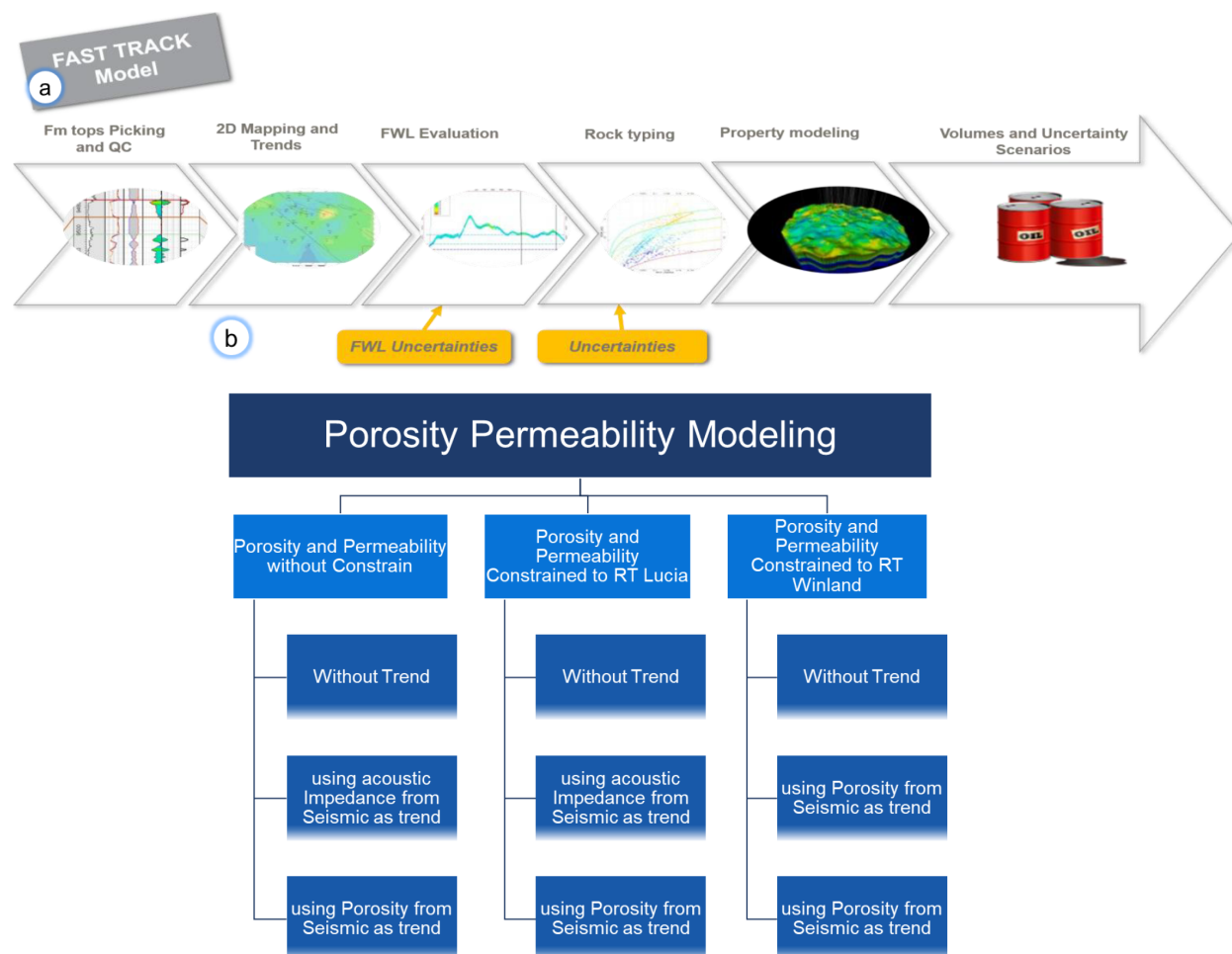


Figure 7—(a) showing the fast track modeling workflow; (b) different Petrophysical Modelig realizations approaches

Extracted Porosity map from after property Modeling is shown in below Fig. 8b compared to the 2D well interpolation porosity map in Fig 8a



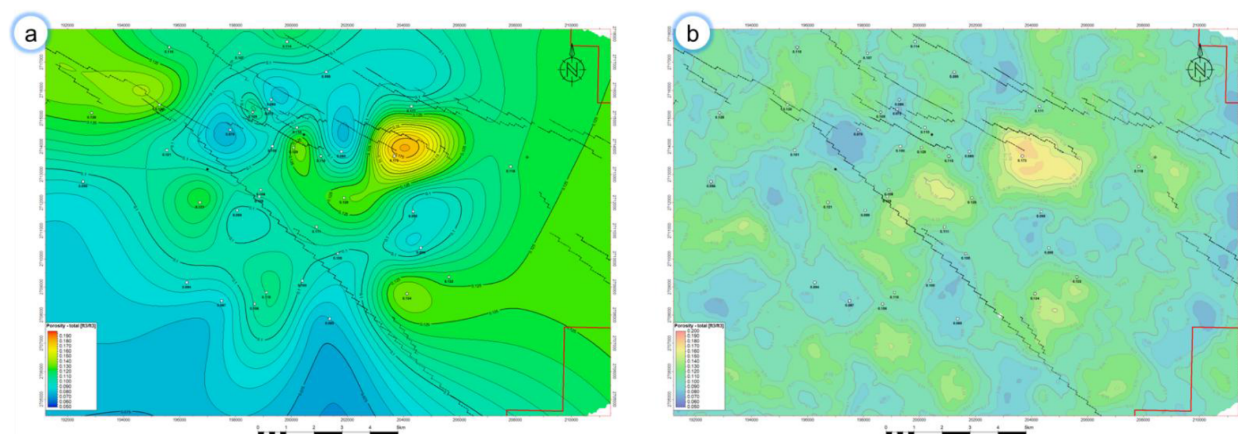
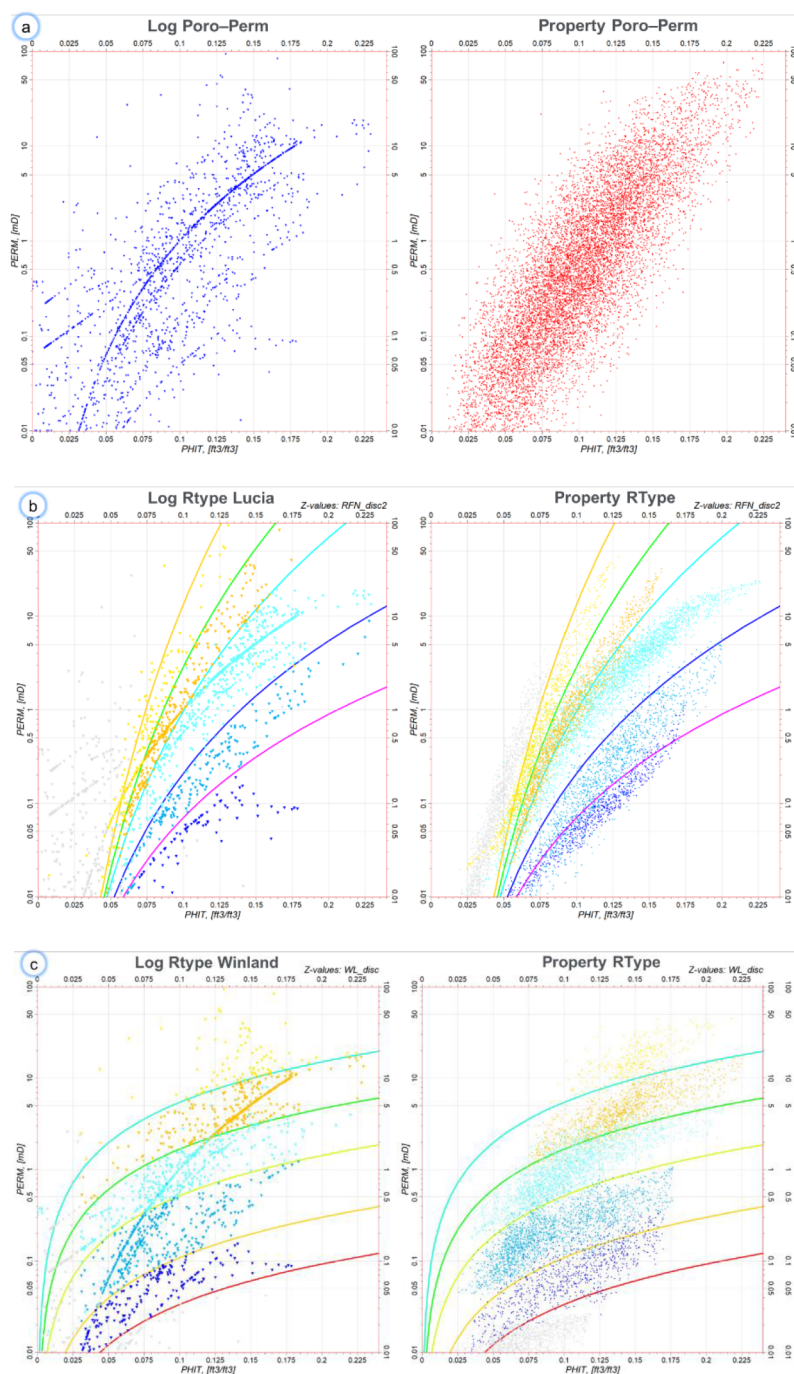


Figure 8—Porosity 2D maps; (a) 2d Porosity map from well data interpolation only; (b) Porosity map extracted from model using Acoustic impedance as a trend

### Geological Model QC

Quality control (QC) on the property model was carried out by comparing the distribution of porosity and permeability from the well data with those in the 3D models. In most of the model versions, or realizations, there was a very good match between the values from the input data and the ones predicted by the model. This provided greater confidence in the modeling results. The comparison process is shown in Figure 9, which highlights how closely the model reflects the actual measurements from the wells. By making these checks, we could be sure that the model was accurate and reliable for volumes calculation.



**Figure 9—Qc on Porosity and Permeability model; 1(a) comparison between porosity and permeability model with well data without using any RT constraining while distribution; (b) comparison between porosity and permeability model with well data using Lucia RT approach; (c) comparison between porosity and permeability model with well data using Winland RT approach**

## Volumes estimation

Ending the geological evaluation, Volumes estimation was calculated for different segments using existing uncertainties which are mainly the FWL uncertainty and Porosity distribution uncertainty.

The output ranges varied from P50 case by +/- 10-15 %, mainly impacted by the property distribution uncertainty.

After finishing the geological study, the team calculated how much hydrocarbon might be in different parts of the reservoir. we did this by looking at the uncertainties that exist right now, such as how unsure we are about the position of the oil-water contact and how porosity is spread out. These uncertainties have

a big effect on the results. The estimates hydrocarbon in place that can range by about 10-15% up or down from the middle case, mostly because of porosity distribution uncertainty.

By understanding these changes and the reasons behind them, the team is better able to plan for the future. This careful way of working means there is less risk when deciding how to develop the field, and it helps make the next steps clearer and more reliable.

### **Way forward**

More detailed analysis would be done after acquiring more Geological and Fluid data from future wells, to have more accurate distribution and volumes estimation with reduced uncertainty, then this would be followed by economical analysis to investigate what could be the optimum way to develop this field economically.

Looking ahead, the team plans to carry out even more detailed studies once new geological and fluid data are collected from future wells. By gathering more information, they hope to build an even clearer picture of how porosity and other important features are distributed throughout the reservoir. This will help them estimate the amount of oil and gas more accurately, and the uncertainty in those numbers will be smaller.

Once the team has better understanding and more reliable volume estimates, they will also do a financial analysis. This step is important because it allows them to look at different ways of developing the field and figure out which plan makes the most sense financially. By following this careful process—collecting more data, improving the models, and studying the costs, the team can make smarter decisions. This approach helps reduce risks and makes the next steps for developing the field much clearer, increasing the chances of a successful and profitable project.

### **Conclusion**

The average gross thickness is approximately 14 feet, with an average net thickness of 12 feet, indicating a tight reservoir. The porosity ranges from 9-13% with average permeability ranging from 1 to 7 millidarcy. The porosity is developed in the crest and northeast area of the field, while it deteriorates in the flanks. The free Water Level (FWL) ranges from 9536 feet to 9614 feet in segments B and D, with possible shallower FWL in segments D and Logs. The rock typing approaches (Lucia and Winland) and permeability from NMR were used. Property modeling considered seismic acoustic impedance and porosity trends. The saturation function from other reservoirs was used in the studied Minor reservoir saturation modeling. The main uncertainty comes from FWL and property distribution. The volume by +/- 10-15 %,